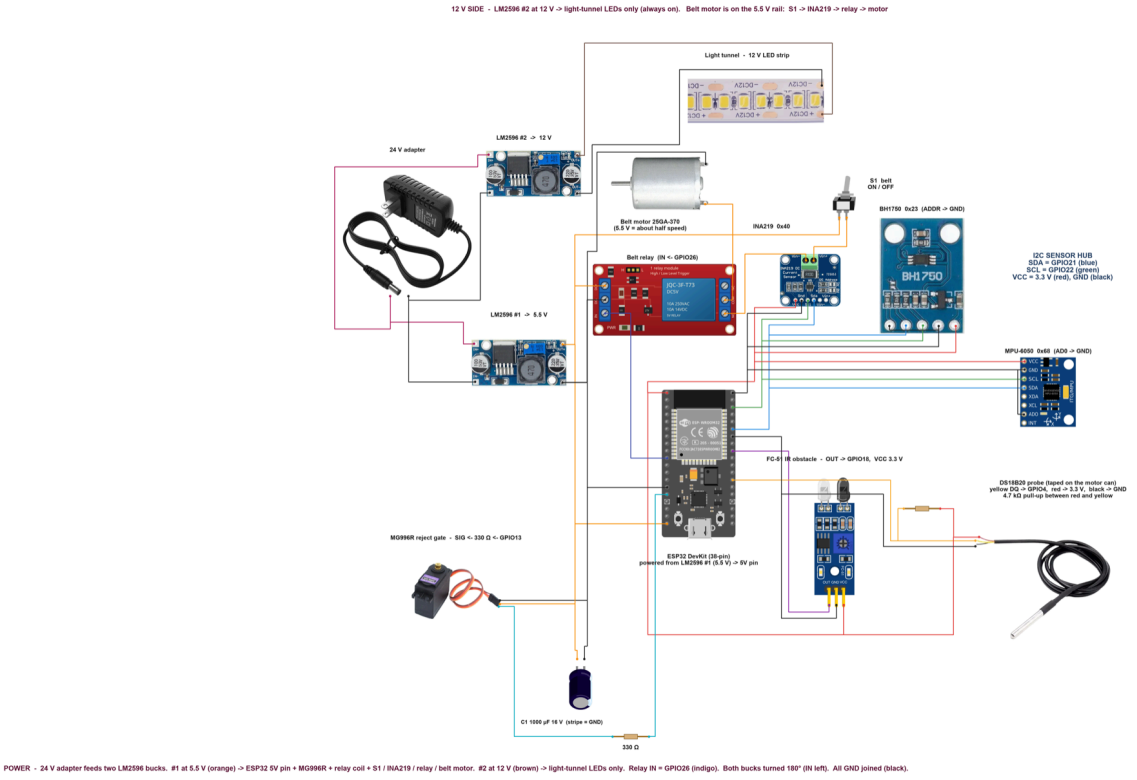
Hardware & Bring-up Guide

The NAZAR rig: power, wiring, sensors, conveyor and gate — and the order to switch it on and test it today.

What the rig is

A short wooden conveyor carries steel washers through a **light tunnel**. A camera looks straight down into the tunnel. When a washer passes the **IR sensor**, the ESP32 lets the belt run a few more centimetres, then **stops it with a relay** so the washer sits still under the camera. The Mac takes the photo, the AI decides PASS / HOLD / REJECT, and the belt restarts. A rejected washer is **kicked off the belt by a servo arm** further down.

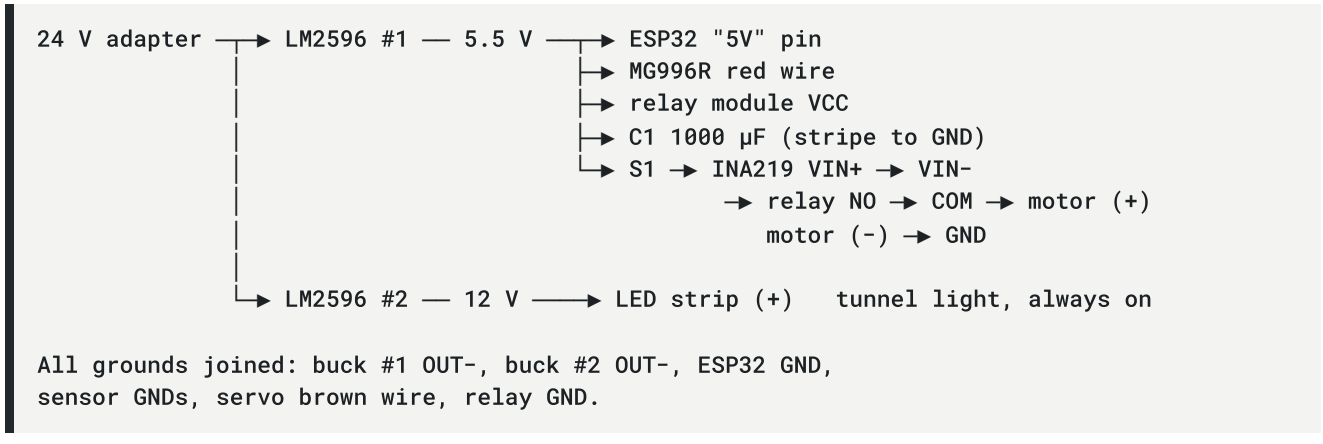
While this happens the ESP32 measures the machine itself: **motor current and voltage** (INA219), **vibration** (MPU-6050), **tunnel light** (BH1750) and **motor temperature** (DS18B20). Those readings are what let NAZAR explain *why* defects happen, not only *that* they happen.



Parts list

Part	Job on the rig	Talks to the ESP32 by
ESP32 DevKitC 38-pin (WROOM, CP2102 USB)	Brain of the line: reads sensors, runs the belt, fires the gate, streams data to the Mac	USB serial 115200
24 V DC adapter	Main power in	—
LM2596 buck #1, set to **5.5 V**	Main rail: ESP32 5V pin, MG996R servo, relay coil, belt motor	—
LM2596 buck #2, set to **12 V**	LED rail: light-tunnel LED strip only	—
1000 µF 16 V capacitor (C1)	Absorbs the servo's current kick on the 5.5 V rail	—
25GA-370 12 V gear motor (run at 5.5 V)	Drives the belt through 18T/36T printed gears	via relay
1-channel 5 V relay module (BC547 driver)	Starts / stops the belt (stop-and-inspect)	GPIO26, active HIGH
Toggle switch S1	Master belt switch (the real "off")	—
INA219 current sensor	Belt-motor current (mA) and voltage (V)	I^2C 0x40
MPU-6050 accelerometer	Motor / frame vibration	I^2C 0x68
BH1750 light sensor	Light level inside the tunnel	I^2C 0x23
DS18B20 waterproof probe	Motor-can temperature	1-Wire, GPIO4
FC-51 IR obstacle sensor	Detects each washer arriving	GPIO18
MG996R servo + printed arm	Kicks rejected washers off the belt	PWM, GPIO13
12 V white LED strip	Even light in the tunnel, always on	—
Resistors: 4.7 kΩ, 330 Ω	DS18B20 pull-up; servo signal series resistor	—
Camera (Logitech C270 / phone via Iriun)	Top-down photo of each washer	USB to the Mac

Power — two rails from one adapter



Why two bucks? The 12 V LED strip needs 12 V; everything else runs on 5.5 V. The belt motor is a 12 V motor, but it runs on the 5.5 V rail at about half speed (roughly 60 rpm instead of 130) — slow is fine, even good, for stop-and-inspect. Buck #1 now carries the ESP32, servo, relay and motor, so give it a heatsink and short thick wires.

Why 5.5 V and not 5.0 V? The MG996R is stronger above 5 V (rated 4.8–7.2 V), and the ESP32's own regulator takes 5.5 V easily. Do not go above 6 V on rail #1.

Why the INA219 sits before the relay: it measures the current the motor actually draws. A jammed belt shows as high current; a slipping or stopped belt shows as low current.

Setting the bucks — do this before connecting anything

1. Connect only the 24 V adapter to a buck's IN+/IN−. Nothing on OUT.
2. Put a multimeter on OUT+/OUT−. Turn the brass trimmer screw (many turns; clockwise usually lowers the voltage) until it reads **5.5 V** on #1, **12.0 V** on #2.
3. Mark each buck with tape: "5.5 V" and "12 V". Swapping them would kill the ESP32.

ESP32 pin map

Signal	ESP32 GPIO	Wire colour (diagram)	Notes
I²C SDA	21	blue	shared by INA219, MPU-6050, BH1750
I²C SCL	22	green	shared
Sensor power	3V3	red	all four sensors run on 3.3 V
FC-51 OUT	18	purple	LOW when a washer is in front
DS18B20 DQ (yellow)	4	yellow	4.7 kΩ between yellow and red
MG996R signal (orange)	13	cyan	through 330 Ω
Relay IN	26	indigo	HIGH = belt runs; LOW = belt stops (active HIGH board)
Power in	5V	orange	from buck #1 (5.5 V)
GND	GND	black	common ground

I²C addresses: INA219 **0x40**, MPU-6050 **0x68** (AD0 to GND), BH1750 **0x23** (ADDR to GND). No two are the same, so they can share the two wires.

How each sensor works (the short version)

INA219 — current and voltage. A tiny 0.1 Ω shunt resistor sits in the motor's + wire. Current through it makes a small voltage drop; the INA219 measures that drop with a 12-bit converter and reports current.

It also measures the bus voltage. On this rig: idle belt about 100–300 mA; a jam pushes it towards 700 mA+.

MPU-6050 — vibration. A MEMS accelerometer: a microscopic mass on springs inside the chip; acceleration moves it, which changes a capacitance. The firmware reads it 200 times a second, takes the size of the acceleration vector |a|, subtracts its slow average (gravity and tilt), and reports the RMS and peak of what is left — that is the shaking. The team's board is mounted tilted about 77°; using |a| makes that irrelevant. Quiet motor: about 0.002–0.02 g. A loose motor mount: 0.08 g and up.

BH1750 — light. A photodiode with a filter shaped like the human eye's response, plus an ADC that gives lux directly. If the tunnel light drops, the camera picture changes and the AI gets less reliable, so NAZAR warns before that happens.

DS18B20 — temperature. A digital thermometer on a single data wire ("1-Wire"). The 4.7 kΩ pull-up keeps that wire HIGH when nobody is talking. A conversion takes 750 ms, so the firmware starts one, does other work, and collects the result later — the 10 Hz data stream never pauses.

FC-51 — part detector. An IR LED shines out; if something reflects it back into the photodiode, a comparator pulls OUT LOW. The blue trimmer sets the distance. Set it so a washer on the belt triggers it and the empty belt does not.

MG996R — reject gate. A servo turns to an angle set by pulse width (about 0.5–2.5 ms, every 20 ms). The ESP32 makes that signal with its LEDC PWM hardware. The 330 Ω resistor protects the GPIO if the servo misbehaves.

Relay — belt start/stop. Our board drives the 5 V relay coil with a BC547 (NPN) transistor, so its input is *active HIGH*: the ESP32 sets IN to 3.3 V to switch the relay on and to 0 V to switch it off. The red LED shows when the relay is on. The diagram draws a different-looking module (with an H/L jumper); the three pins are the same — VCC, GND, IN. Its screw terminals are not labelled on the front: with the relay off, COM and NC beep on a multimeter; the third one is NO.

Wiring and bring-up order

Wire one block, test it with its bring-up sketch from **nazar-firmware/bringup/**, then move on. If a test fails, you know it is the last block you touched.

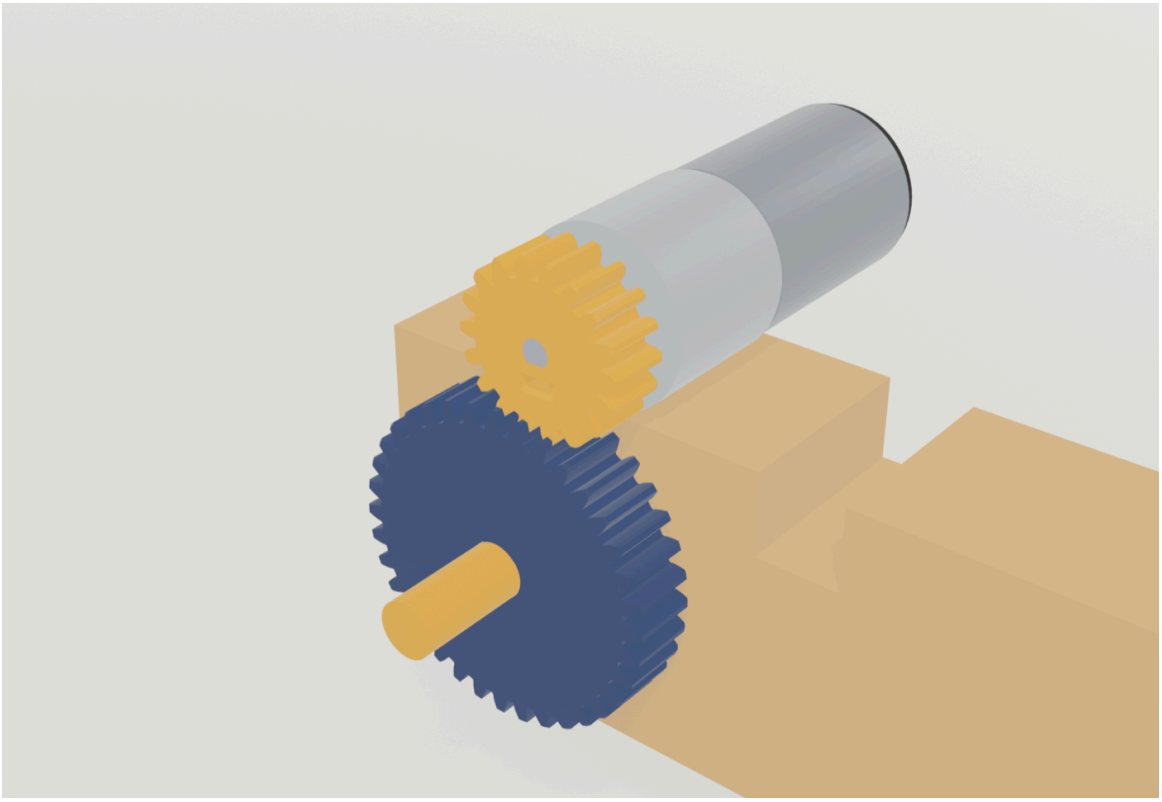
Step	Wire this	Flash and check	It passes when
0	Nothing. ESP32 on laptop USB only	Any sketch uploads	Upload finishes; port is `/dev/cu.usbserial-…`
1	INA219, MPU-6050, BH1750 on 3V3/GND/21/22	`01_i2c_scan`	Finds 0x23, 0x40, 0x68 — exactly these three
2	—	`02_ina219`, `03_mpu6050`, `04_bh1750`	Current ≈ 0 with motor off; vibration rises when you tap the board; lux drops when you cover the sensor
3	DS18B20 + 4.7 kΩ on GPIO4	`05_ds18b20`	Room temperature; rises when you hold the tip
4	FC-51 on 3V3/GND/18	`06_fc51_ir`	Prints PART when a washer passes, CLEAR otherwise; adjust trimmer
5	Set bucks (above). Wire buck #1 → ESP32 5V, C1, servo, relay VCC	`07_servo_gate`	Arm moves between parked and kick angles; ESP32 does not reset during the move
6	Buck #2 → LEDs; buck #1 → S1 → INA219 → relay → motor	`08_relay_belt`	`1` runs the belt, `0` stops it; S1 off always stops it
7	Everything	`nazar_line` (main firmware) + `tools/serial_check.py`	`hello` lists all four sensors true; telemetry 10 lines/s; PING → pong
8	Camera + Mac app	NAZAR Console	LIVE DATA band; a washer stops under the camera; verdict comes back; REJECT kicks it off

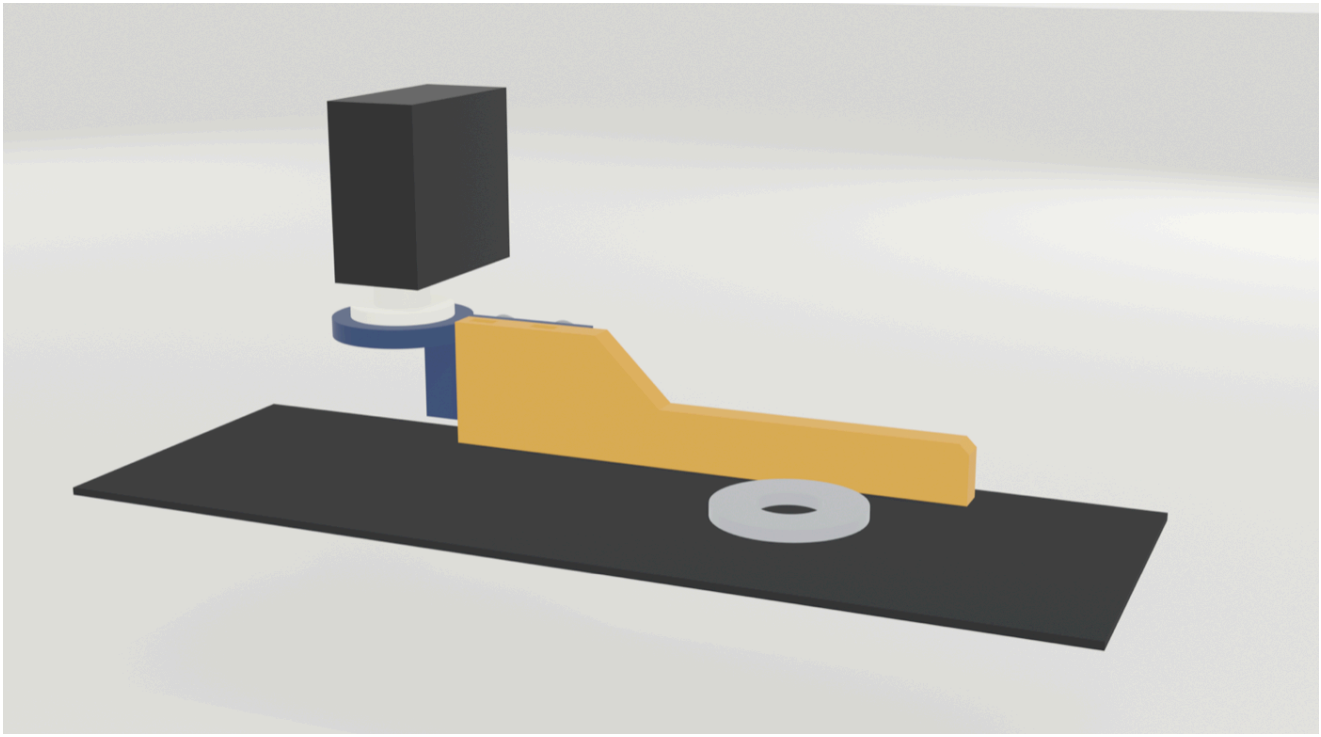
Before step 5: unplug USB, power up from the 24 V side once and measure 5.5 V at the ESP32 5V pin. Most DevKitC boards have a diode so USB and the 5V pin can be connected together; if you are not sure, run from one source at a time.

Conveyor mechanics

- **Frame:** two wooden rails 640 × 80 × 18 mm, 135 mm apart inside, on leg blocks.
- **Rollers:** 25 mm PVC pipe, 125 mm long, with 3D-printed end plugs on 608ZZ bearings; 540 mm centre to centre.
- **Drive:** the 25GA-370 (12 V, 130 rpm; about 60 rpm on our 5.5 V rail) lies on top of the rail in the printed **P7S saddle** and drives the roller through printed **18T / 36T gears** (2:1, 33 mm centre distance). Slide the motor along the rail to set the mesh. Gears reverse direction compared with a belt — swap the motor wires if the belt runs backwards.
- **Light tunnel:** printed brackets hold the 12 V LED strip; the camera looks down through the top. Keep camera height and lighting **the same as when the AI was trained**.
- **Reject gate:** MG996R on the printed **P13 tower**, with the **P14 sort arm** (15 mm height adjustment). It is a *kick*, not a slow diverter: parked along the rail, it swings about 70°, holds ~0.35 s and returns.

Washers need about 100 mm spacing.





Printable files for these parts are in `mechanical/stl/`.

Camera setup for good AI results

1. Camera straight down, fixed height, washer filling about a third of the frame.

2. Tunnel LEDs on. In NAZAR Console open **Camera → Settings**: switch off automatic exposure, lower exposure until the "too bright" meter reads near 0 % with a washer in view (on the C270 about 1.5 ms, gain 0), set power-line frequency to **50 Hz**.
3. Use the console's capture mode to photograph 40–50 good washers in different positions, then retrain (see the Vision AI guide).

Troubleshooting

Symptom	Likely cause	Fix
I²C scan finds nothing	SDA/SCL swapped, or no 3V3	Blue = 21, green = 22; measure 3.3 V at each sensor
MPU answers 0x70/0x72/0x98 instead of 0x68	Clone chip	Fine — firmware accepts it
DS18B20 reads –127 °C or 85 °C	Missing pull-up / read too early	4.7 kΩ between yellow and red
ESP32 resets when the servo moves	5.5 V rail sags	C1 fitted, short thick wires, buck #1 not overloaded
ESP32 resets when the belt starts	motor start current sags the 5.5 V rail (the motor shares it now)	same fixes, plus 470–1000 µF right at the ESP32 5V/GND pins
Belt never stops	Motor on NC instead of NO, or IN wired to the wrong pin	Motor + through NO → COM, IN to GPIO26
Belt runs when it should stop	Firmware set for an active-LOW module	`RELAY_ACTIVE_HIGH = true` in `config.h`
Belt never runs	S1 off, or relay NO/COM swapped	Motor + goes through COM → NO
Current reads negative	INA219 VIN+ and VIN– swapped	VIN+ from S1 side, VIN– to relay
Upload fails "port busy"	Console app or serial monitor holds the port	Disconnect in NAZAR Console first
AI gives HOLD on good washers	Photo too bright or light changed	Lower exposure, recapture, retrain

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Safety

- S1 is the master belt switch. Keep fingers clear of the gears when S1 is on — the relay can restart the belt at any moment.
- The bucks get hot under load; buck #1 needs its heatsink.
- Never connect the 24 V adapter directly to anything except the buck inputs.

Status today (2 Nov, onsite)

- **Designed and drawn:** full wiring (Cirkit Designer files in `cirkit/`, renders and PDF in `diagrams/`), mechanical parts printed.
- **Tested on a real board earlier (1 Nov):** ESP32 enumerates; MPU-6050 genuine at 0x68.
- **Not yet tested on the real rig:** the other sensors, relay, servo kick timing, full stop-and-inspect cycle. That is today's bring-up, in the order above.